Aaryan Samanta

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Summary

- **Research Scholar (Mentor: Christos E. Constantinou, Stanford University School of Medicine Associate Professor of Urology, Emeritus),** Jan'26–Present: Analyzed anonymized pelvic floor ultrasound data across cohorts.
- **Lead author, two peer-reviewed publications:** IEEE AIAM 2025 (quantum-inspired GA-GNN ensemble) and IJHSR (C. elegans mitochondrial resilience); rare achievement for high schoolers in competitive venues.
- **USACO** (USA Computing Olympiad) **Platinum Division (Certified): Triple perfect scores** (1000/1000 Bronze/Silver/Gold); top ~2-5% globally
- **USABO** (USA Biology Olympiad) **Semi-Finalist 2026:** 43/50 on Open Exam (elite score, well above cutoff; top ~0.5% nationally)
- **AMC 10 & 12 National 1st Place Perfect Scores** (150/150); 2× AIME qualifier
- **Founder & President** of **AI Ethos, Inc**. (501(c)(3)) and **NextGenAI International**; scaled ethical AI education to 75+ students across 3 countries.
- SAT **1570** (800 Math) | GPA **4.8 W / 4.0 UW** | UCSB Summer Research Academy **youngest scholar**

Areas of Interest

- Artificial Intelligence
- Machine Learning
- Graph Neural Networks
- Ethical AI
- Quantum Computing
- Competitive Algorithms
- Bioinformatics & Med Imaging
- Healthcare AI

Education

- **Legend College Preparatory**, Cupertino, CA, 10th Grade, Class of 2028
- **GPA: 4.80** Weighted, 4.00 Unweighted
- **AP Scores: 5 on all 3** completed (**planned: 25** total by graduation)
- **SAT: 1570** (**800 Math**)

Research Experience & Peer-Reviewed Publications

- **Research Scholar (Mentor: Christos E. Constantinou, Stanford University School of Medicine Associate Professor of Urology, Emeritus,), Jan'26–Present**: Analyzed anonymized pelvic floor ultrasound data across cohorts & stimulation states; developed 3D models from data for kinematic & pathophysiologic insights. Link
- **Lead Author, IEEE Artificial Intelligence and Advanced Manufacturing (AIAM) 2025 International Conference Publication, Nov'25**: *Quantum-Inspired Hybrid Genetic Algorithm and Graph Neural Network Ensemble for Multimodal Classification*; peer-reviewed IEEE proceedings. Link
- **University of California, Santa Barbara (UCSB) Summer Research Academy (SRA) Scholar & Lead Author, International Journal of High School Research** (**IJHSR**) **Pub. Jul–Dec'25:** Youngest accepted (top ~10% rate); capstone "Resetting the Clocks: Mitochondrial Quality Control via Genetic Adaptation"—identified resilience genes in C. elegans CX11314/EG4725 via comp bio analysis; paper: "Natural genetic variation in mitochondrial health-regulating genes..."; peer-reviewed journal. Link
- **AI Research Intern, Legend College Prep AI Lab**, **Jun'25:** Built CNN for pneumonia detection from chest X-rays (>90% accuracy); developed full pipeline (preprocessing, augmentation, training, evaluation). Link
- **Data Science Research Scholar, Legend College Prep AI Lab, Sep '25**: Disease-Symptom Demographic EDA; applied ML/statistical analysis to evaluate patient data, identify demographic correlations, and built public-health visualizations. Link

Leadership & Global Impact

- **Founder & President, AI Ethos, Inc. (501(c)(3))** • 2025–Present: Directed ethical AI bias mitigation workshops for 75+ students across 3 countries; promoted equitable AI access. http://aiethos.org/
- **Founder & President, NextGenAI International** • 2025–Present: Built international student organization with 50+ members; hosted international seminars expanding global student network. http://nextgenai.org
- **Founder/Chief**, **Global AI Ambassador** • 2025–Present: Created first ever high school Global AI ambassador role; collaborated with educators and principals in Taiwan and the U.S. to advance responsible AI adoption.
- **Co-Captain, Cupertino Junior Olympiad Math/CS Team** • 2024–Present: Mentored 20+ peers to AIME/USACO Gold qualifications.
- **Volunteer Team Leader, VA Palo Alto Health Care** • 2024–Present: Directed audits of 50+ elderly medical images; standardized AI-assisted diagnostic workflows.

Awards & Honors

- **Lead Author, IEEE AIAM 2025** Peer-Reviewed Publication (International AI & Mfg Conference, <1% HS acceptance)
- **USACO** (USA Computing Olympiad) **Platinum Division (Certified): Triple perfect scores** (1000/1000 Bronze/Silver/Gold); top ~2-5% globally
- **USABO** (USA Biology Olympiad) **Semi-Finalist 2026:** 43/50 on Open Exam (elite score, well above cutoff; top ~0.5% nationally)
- **AMC 10 & 12 National 1st Place** Perfect Scores (150/150); 2× AIME Qualifier, 2024-2025
- **UCSB Summer Research Academy Scholar** – Youngest accepted (top ~10 % admission rate, IJHSR publication).
- **Lead Author, IJHSR,** Peer-Reviewed (International Journal of High School Research)
- **NSHSS National Scholar** (Top 1% Academic & Leadership Recognition).

Additional Distinctions

- **Track & Field JV Anchor**, MVHS Matadors: Set all-time 9th grade school records (100m: 11.75s); Rookie of the Year; League Champions Link
- **Percussion Section Leader**, MVHS Concert Band: Directed 10-member section to Unanimous Superior & Perfect Score, CMEA Festival.

Technical Proficiency

- **Domains:** Graph Neural Networks, Genetic Algorithms, CNNs, Bioinformatics, Ethical AI
- **Languages:** Python, PyTorch, TensorFlow, NumPy, Pandas, scikit-learn), Java, JavaScript
- **Tools:** MATLAB, Git, LaTeX, Linux

References

- Dr. Tian Yishuai, Postdoctoral Fellow, Tsinghua University & Research Fellow, Nanjing University
- Dr. Li Chieh Lin, Chairperson, Taiwan Artificial Intelligence Industry Association (TAIA)
- Laurence Dang, Dean of Academics, Chair, Dept. of Mathematics and Sciences, Legend College Preparatory
- Carlos Febrero, AI Expert & Mentor, Computer Science Instructor, Legend College Preparatory

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Profile & Links

- **Personal Website:** aaryansamanta.com
- **GitHub:** github.com/aaryansamanta
- **IEEE AIAM Publication:** ieeexplore.ieee.org/document/11322272
- **IJHSR Publication:** Link
- **Google Scholar:** scholar.google.com/user=TboJyScAAAAJ, **ORCID:** orcid.org/0009-0005-6381-4890
- **AI Ethos Non-Profit:** aiethos.org
- **NextGenAI Organization:** nextgenai.org, **Videos:** 1, 2
- **Track & Field Athletic Profile:** NCSA, Matadors

World Language Proficiency

- **Spanish:** AP Spanish Level
- **Bengali:** Beginner Level